

Outset · AI in Healthcare · Day 3 · Group 1

# Picking good CRISPR guides before the lab

Given the 20 letters of a guide RNA, can a model tell -- before any lab work -- whether CRISPR-Cas9 will actually cut?

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**Outset**

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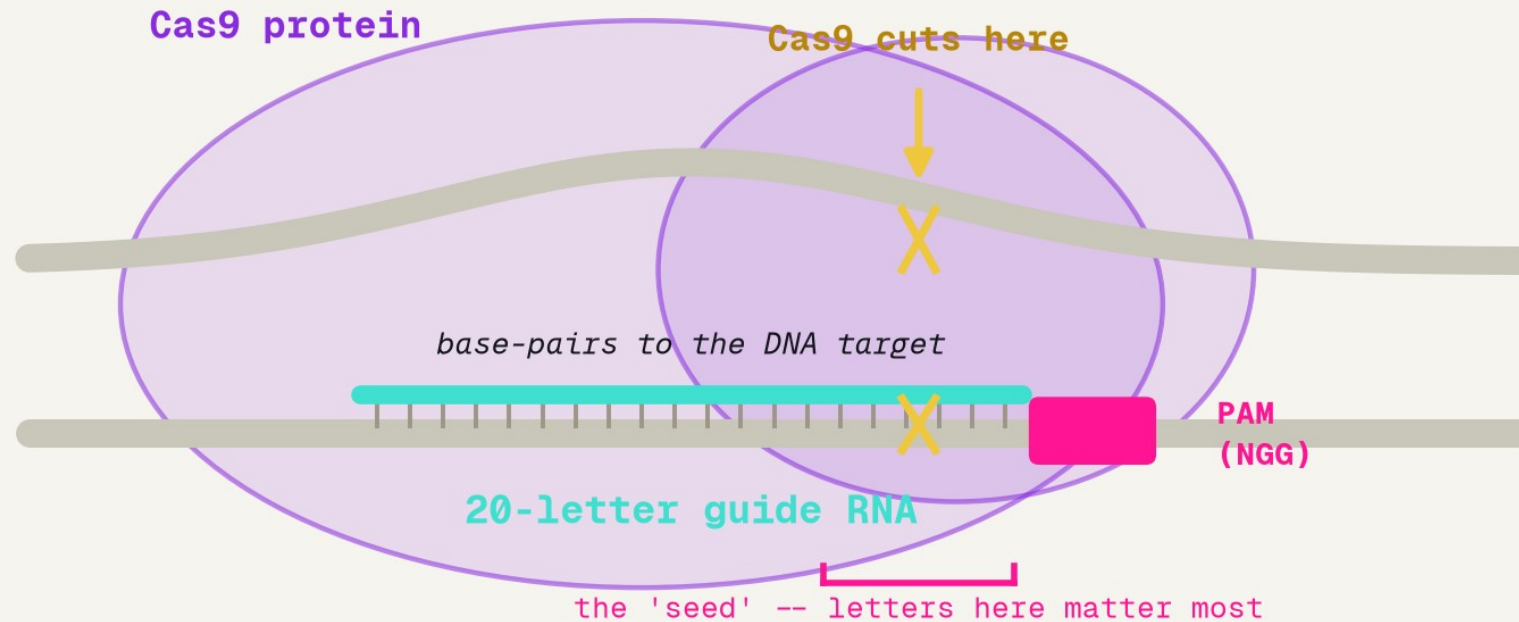
2026-07-08

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# BACKGROUND

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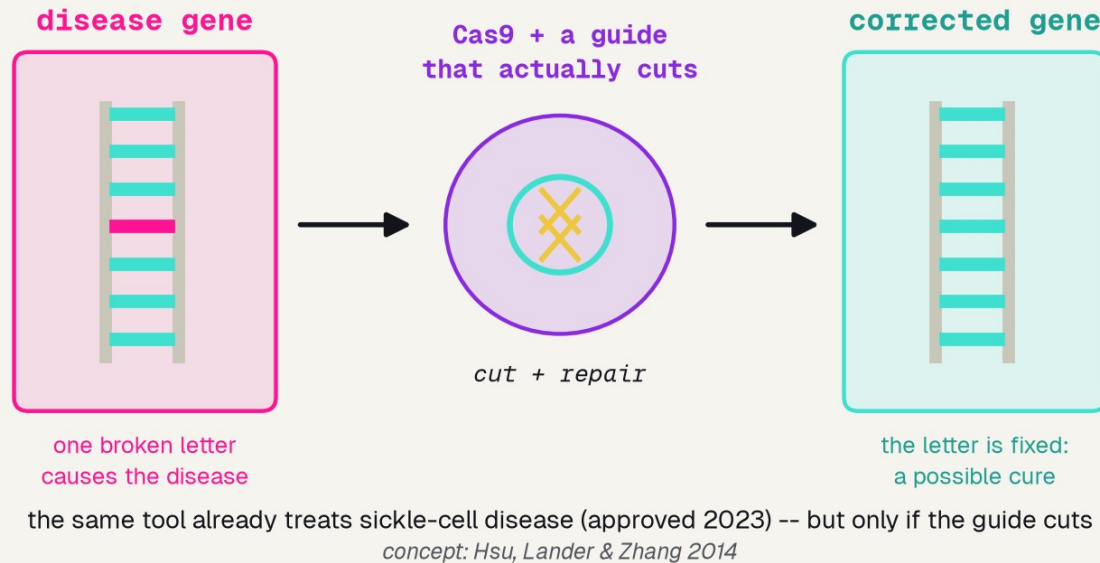
# Molecular scissors, aimed by 20 letters



*adapted from Hsu, Lander & Zhang 2014*

*A 20-letter guide RNA steers Cas9 to one spot in DNA; it cuts about three letters from the PAM -- and the letters in the seed, next to the PAM, matter most.*

# CRISPR as therapy: fixing a broken gene



## THE CLINICAL SETTING

### Who

patients with a genetic disease -- sickle-cell today, many more in trials

### Where

research labs and gene-therapy clinics deciding what to build

### The decision

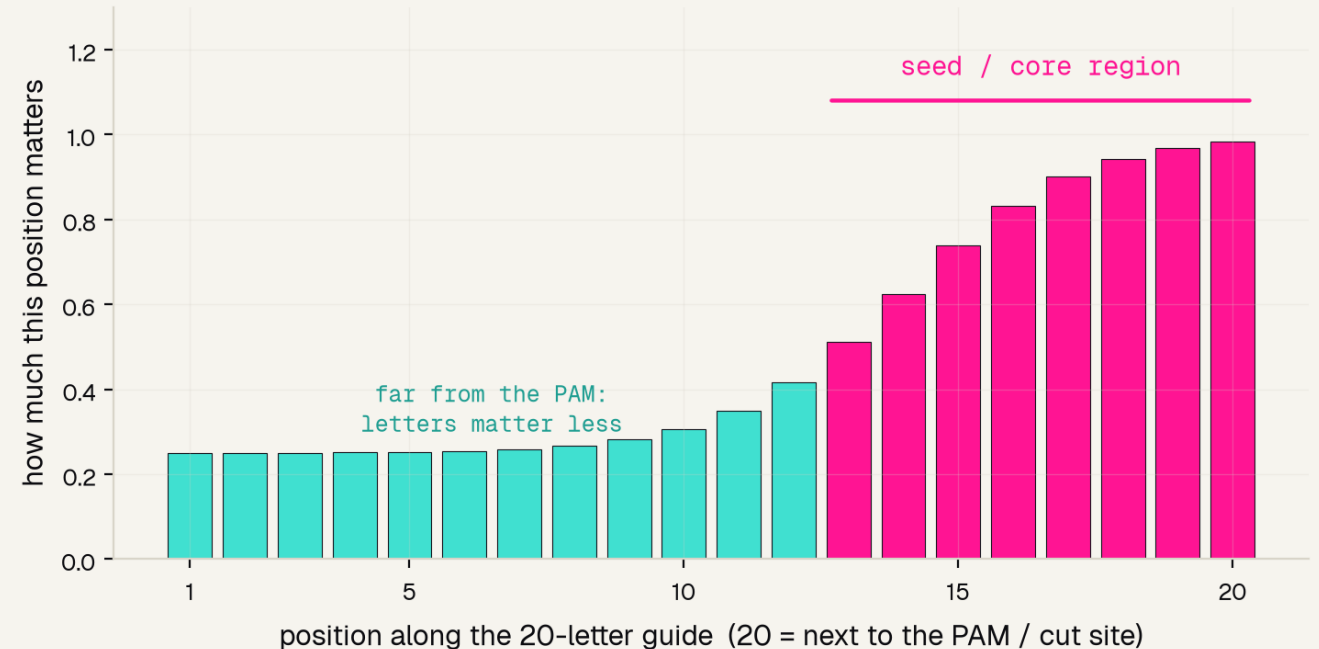
which 20-letter guide to make -- the one that will actually cut

# Not all 20 letters count equally

## THE ONE BIT OF BIOLOGY

Cas9 lands on the PAM, then reads the handful of letters right beside it -- the seed.

Mismatch a seed letter and the guide usually fails to cut. So the letters near the PAM should carry the most signal -- far more than the ones at the far end.

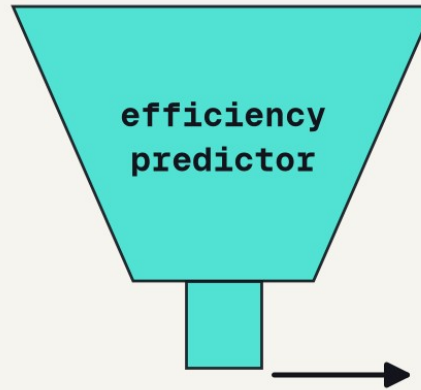
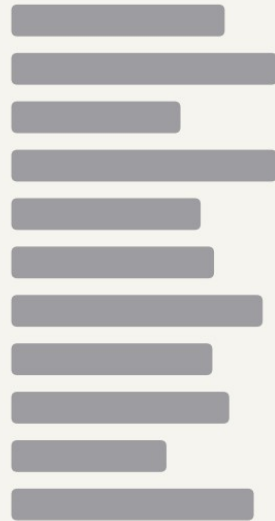


adapted from Zheng et al. 2017 / Doench et al. 2016

*Remember this -- our model will rediscover it.*

# The catch: most candidate guides are duds

dozens of candidate guides  
for a single gene



order only the top few  
the ones most likely to cut

good guide

good guide

good guide

*concept adapted from Doench et al. 2016*

## the old way

test dozens of guides in the lab, one by one -- weeks of work, and most turn out to be duds

## the tool

score all of them from the letters in seconds, then order only the top few most likely to cut

# THE DATA

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# Why Doench 2016

**5,310**

guides, each ACTUALLY tested in the lab -- so every cut-score is a real measurement, not a simulation

**17**

genes targeted, a spread wide enough that the model learns sequence rules, not the quirks of one gene

**the**

benchmark: the dataset Doench built Rule Set 2 / Azimuth on -- what the whole field still compares against

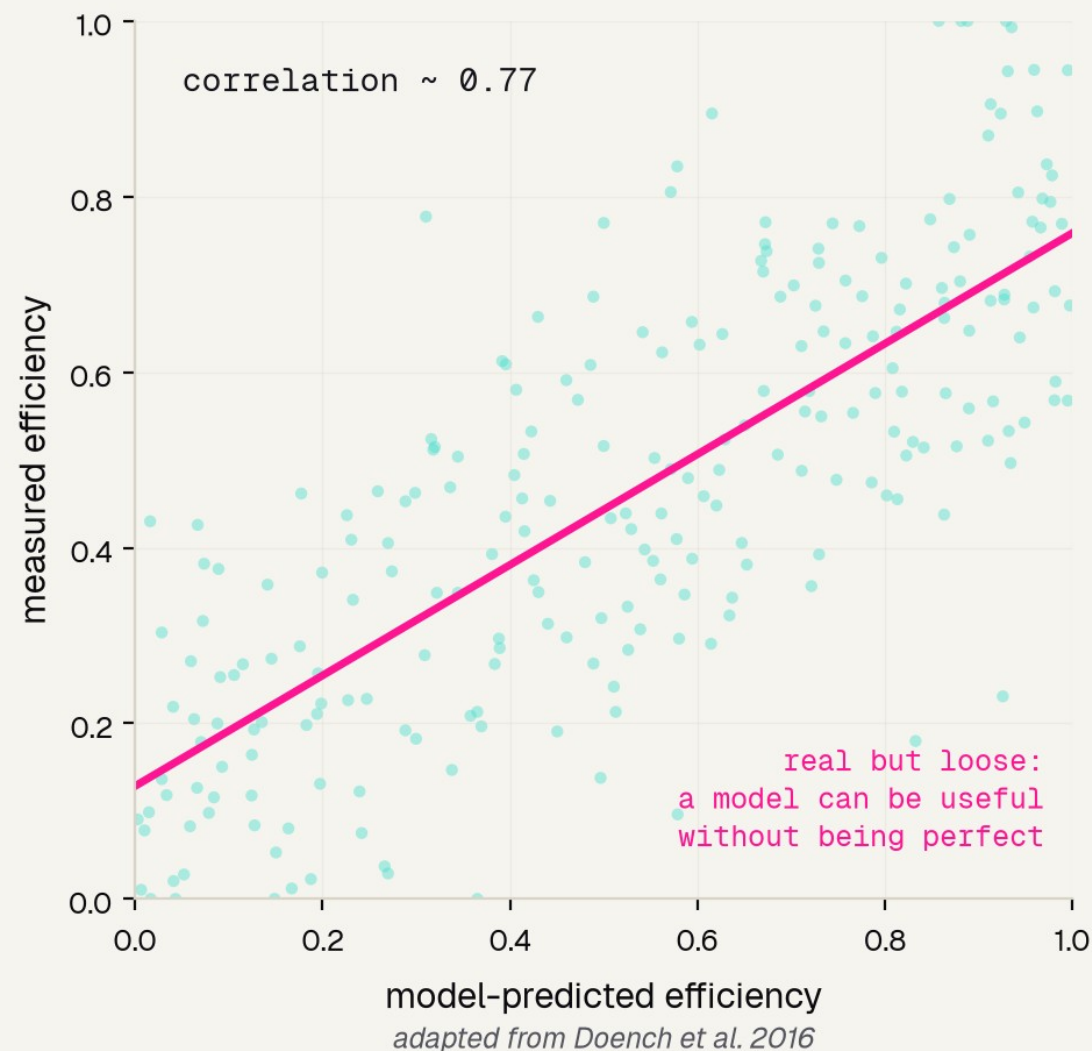
# The benchmark that set the standard

## SET HONEST EXPECTATIONS

Doench 2016 built Rule Set 2 (Azimuth) on exactly these guides -- the state of the art.

Even so, its predicted scores only loosely track the measured ones (the cloud scatters around the line).

*So our goal is to capture the real signal -- not to expect a perfect predictor.*



# THE MODEL

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# First, turn letters into numbers

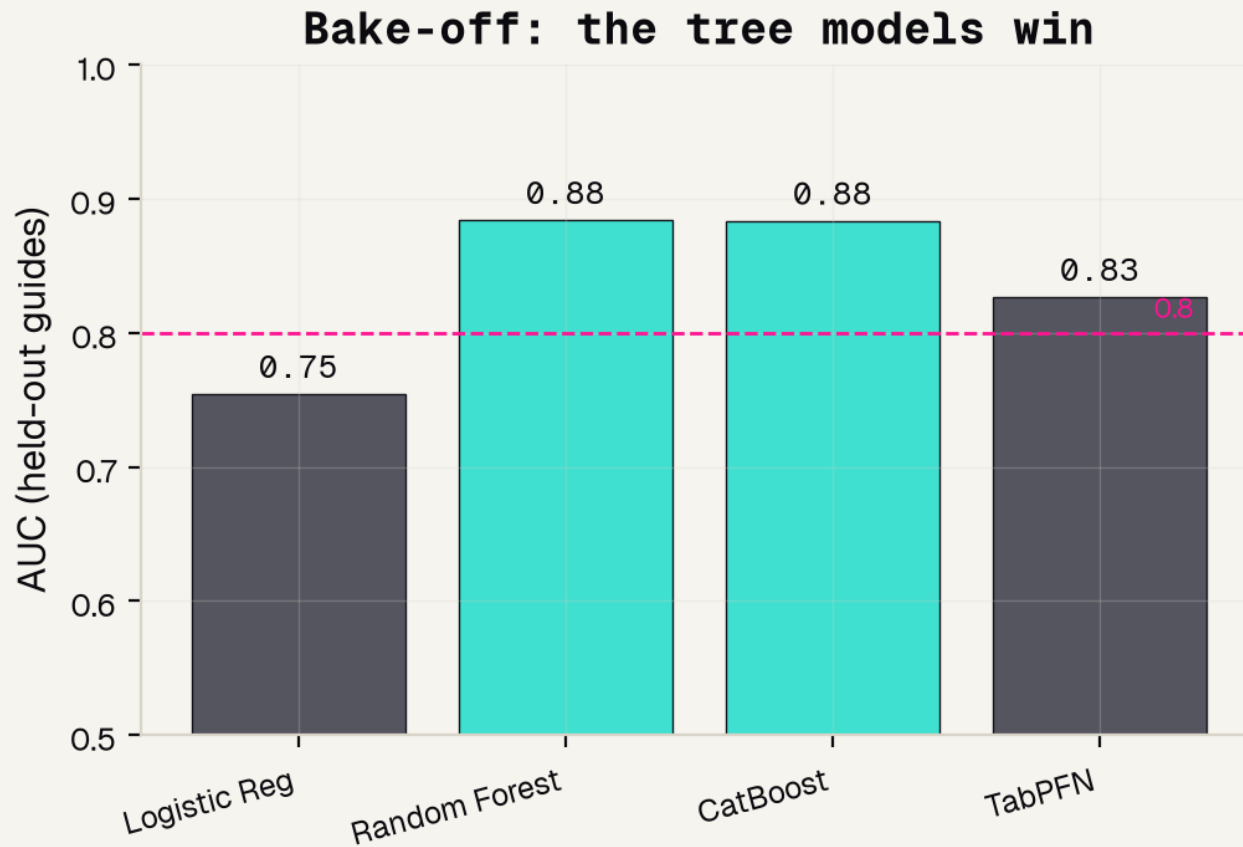
a computer cannot do math on the letter A -- so one-hot flips 4 switches per position:

|   | G | A | C | T | G | A | A | C | A | G |
|---|---|---|---|---|---|---|---|---|---|---|
| A |   | 1 |   |   |   | 1 | 1 |   | 1 |   |
| C |   |   | 1 |   |   |   |   | 1 |   |   |
| G | 1 |   |   |   | 1 |   |   |   |   | 1 |
| T |   |   |   | 1 |   |   |   |   |   |   |

One switch ON per column → 80 numbers that KEEP the order, so the seed near the PAM stays visible.

*k-mer would just count letters and throw the order away -- so we use one-hot.*

# We tested four models - we did not guess



WE TESTED, WE DID NOT GUESS

## 0.88

the two TREE models (CatBoost and Random Forest) tie here -- both beat TabPFN (0.83) and crush logistic regression (0.75).

*We pick CatBoost -- fast, strong, and reproducible.*

# Model and data processing

1

**TASK**

clearly-good (top-third efficiency) vs clearly-poor (bottom-third); the muddy middle is dropped

2

**ENCODE**

each 20-letter guide becomes an 80-number one-hot vector that keeps the position of every letter

3

**SPLIT**

a stratified 75% train / 25% test split -- every number we report is on guides the model never saw

4

**MODEL**

CatBoostClassifier, 600 trees, depth 6, learning rate 0.05, random\_state 0 so it is reproducible

5

**GRADE**

AUC on the held-out test set -- not accuracy -- so a lazy one-class guess cannot game the score

# RESULTS

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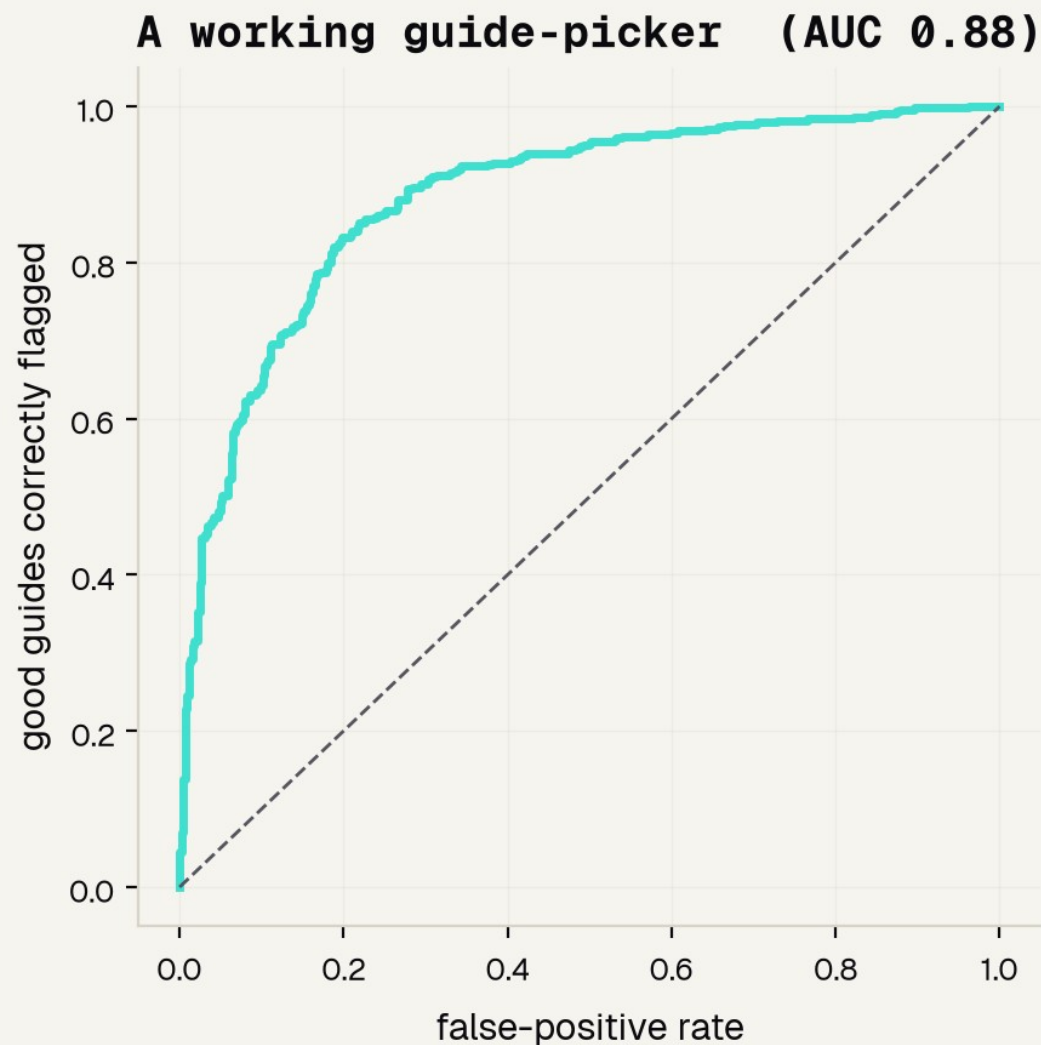
# How we grade it: ROC and AUC

0.88

**AUC** -- the area under the ROC curve

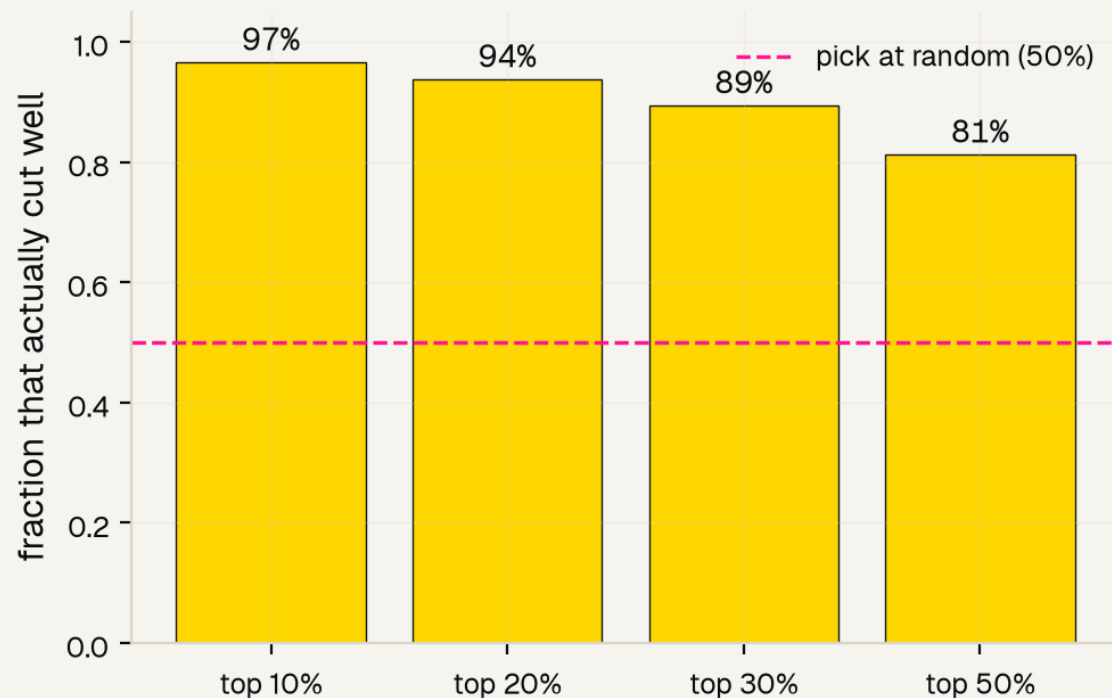
The picker outputs a score, not a yes/no. The ROC curve sweeps every possible cut-off and plots good guides caught against false alarms; AUC is the area underneath.

0.5 = a coin flip      1.0 = perfect



# The win: order its top picks

Order the model's top picks, get mostly good guides



97%

of the model's TOP 10% actually cut well

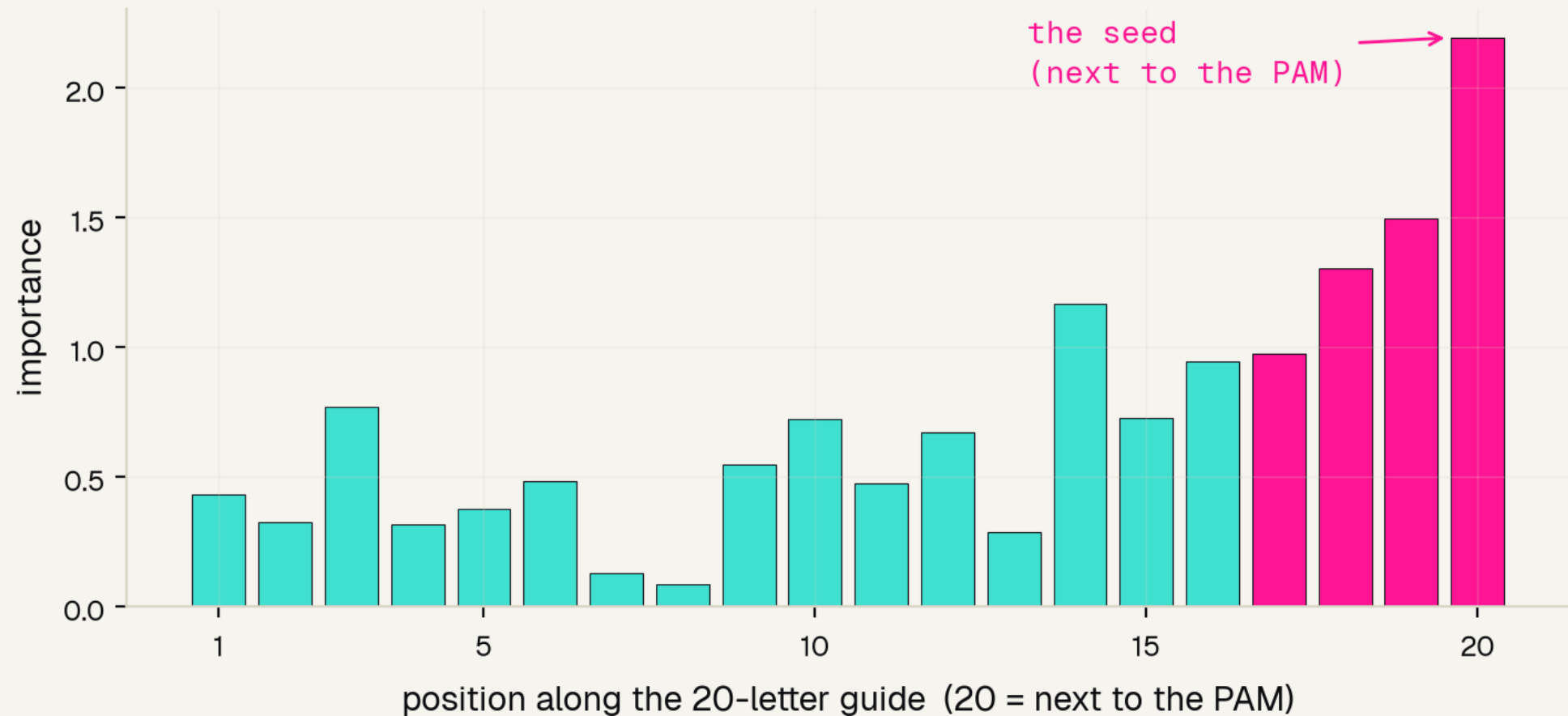
50%

if you pick guides at random -- the baseline

weeks

of lab time saved, because the shortlist is almost all winners

# It rediscovered the biology



Feature importance = which positions the model leaned on. It peaks at position 20 -- right next to the PAM, the seed -- matching decades of CRISPR biology it was never told.

# Fairness? There is nothing to audit - honestly

**No patient. No demographics. Nothing to be unfair about.**

Most medical-AI projects must check they work equally across sex, age, or skin tone. This one has no such groups -- the input is a 20-letter DNA sequence. Forcing a fake fairness analysis would be less honest than simply saying so.

*The real equity question for CRISPR lives one step later -- in which patients a therapy is tested on and reaches -- outside this sequence model.*

# THE BOTTOM LINE

# What it can and cannot do

## IT WORKS

A real guide-picker on the field's benchmark: AUC 0.88, and its top-10% shortlist is 97% good guides -- real lab time saved.

AUC 0.88 · 97% precision

## IT LEARNED REAL BIOLOGY

Importance peaks at the seed next to the PAM, matching decades of CRISPR science -- so we can trust WHY it works.

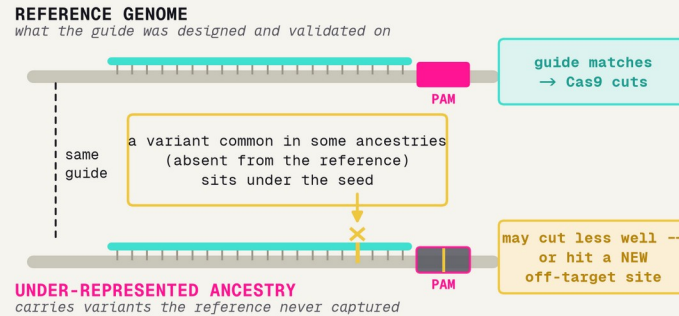
rediscovered the seed

## BUT IT IS A DEMO

It cannot match a production tool, capture cell-type or chromatin effects, or say anything about off-target safety.

not a design tool -- yet

# Where equity lives in genome editing



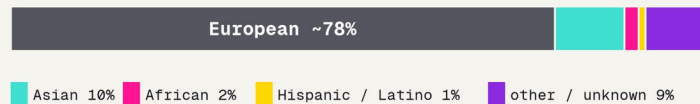
## THE REAL EQUITY QUESTION

Our data is guide sequences -- no patients -- so per-guide fairness does not apply. The real equity issue in genome editing sits one step up.

## THE MECHANISM

Reference genomes and variant databases skew heavily European. A variant common in some ancestries but absent from the reference can sit under the guide or PAM -- a mismatch, an altered PAM, or a new off-target site -- so the same guide may cut less well, or less safely, for those patients.

## WHO IS ACTUALLY IN THE GENOMIC DATABASES



GWAS participants ~78% European (Sirugo et al. 2019); reference + variant panels (gnomAD / 1000 Genomes) skew the same way

adapted from Popejoy & Fullerton 2016 and Sirugo, Williams & Tishkoff 2019

## THE DATA THAT COULD CHECK IT

gnomAD and 1000 Genomes hold ancestry-labeled variant frequencies -- the raw material to flag risky guides before the lab.

[9] Popejoy & Fullerton 2016, Nature -- "Genomics is failing on diversity."

[10] Sirugo, Williams & Tishkoff 2019, Cell -- "The Missing Diversity in Human Genetic Studies."

# References

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## FOUNDATIONS & DATA

- [1] Doench et al. 2016, Nat Biotechnol -- Rule Set 2 / Azimuth, and the dataset we use.
- [2] Doench et al. 2014, Nat Biotechnol -- Rule Set 1: position + PAM predict cutting.
- [3] Hsu, Lander & Zhang 2014, Cell -- how Cas9 is steered to a DNA site (the seed).
- [4] Zheng et al. 2017, Sci Rep -- direct evidence the PAM-proximal seed matters most.

## METHODS & REVIEWS

- [5] Chuai et al. 2018, Genome Biol -- DeepCRISPR: deep learning for guide design.
- [6] Kim et al. 2019, Sci Adv -- DeepSpCas9: more data + neural nets push accuracy further.
- [7] Konstantakos et al. 2022, NAR -- a survey of guide-efficiency predictors.
- [8] Abbaszadeh & Shahlai 2025, arXiv -- explainable AI + off-target safety.

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## EQUITY & DIVERSITY IN GENOMICS

- [9] Popejoy & Fullerton 2016, Nature -- "Genomics is failing on diversity."
- [10] Sirugo, Williams & Tishkoff 2019, Cell -- "The Missing Diversity in Human Genetic Studies."

# The honest bottom line

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**A guide-picker earns its keep by shortlisting the winners -- and explaining itself.**



Use it to build intuition and narrow the field -- then use a validated tool and the lab to actually choose and confirm a guide.

# Thank You

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